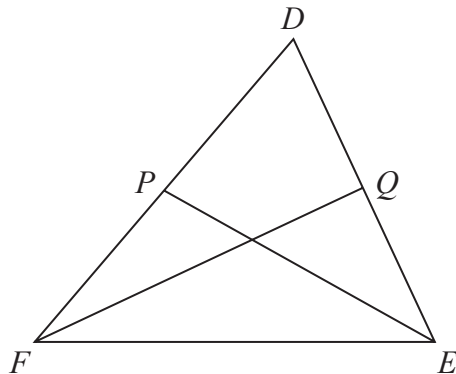


**21**  $DEF$  is a triangle.



$P$  is the midpoint of  $FD$ .

$Q$  is the midpoint of  $DE$ .

$$\overrightarrow{FD} = \mathbf{a} \quad \text{and} \quad \overrightarrow{FE} = \mathbf{b}$$

Use a vector method to prove that  $PQ$  is parallel to  $FE$ .

**(Total for Question 21 is 4 marks)**

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